STM32-BASED RELIABLE LPG GAS LEAKAGE DETECTION SYSTEM

A PROJECT REPORT

submitted by

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CERTIFICATE

This is to certify that the project report entitled **"STM32-Based LPG Gas Leakage Detection System"**, submitted by **ANAMIKA RAVI**, **KRISHNA KRISHNAKUMAR**, **NANDANA DINESHAN P** and **SAFEENA K S** to Muthoot Institute of Technology and Science, Kochi, in partial fulfilment of the requirements for the award of the degree of Bachelor of Technology in Computer Science & Engineering, is a bona fide record of the project work carried out by them under the supervision and guidance of the faculty-in-charge.

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ABSTRACT

Liquefied Petroleum Gas (LPG) is widely used in domestic and commercial settings, but accidental leakage can create serious fire and explosion hazards. This project presents an LPG leakage monitoring prototype built using the STM32F446RE microcontroller and an MQ-6 gas sensor. The sensor output is acquired through the microcontroller's Analog-to-Digital Converter (ADC), and the firmware analyses readings relative to an initial baseline.

The system is designed to consider the sensor level, trend, and persistence of elevated readings when classifying the monitored condition. Three LEDs provide local visual indication: green for Normal, yellow for Warning, and red for Critical. The report describes the hardware and software requirements, system architecture, circuit integration, firmware workflow, testing procedures, and cost estimation. The prototype provides a practical demonstration of embedded sensing and local alerting. Its performance must be established through controlled testing and calibration, and it is not a replacement for a certified LPG safety alarm.

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Chapter 1

INTRODUCTION

Liquefied Petroleum Gas (LPG) is widely used in domestic kitchens, commercial establishments, laboratories, and industrial applications. Although LPG is an efficient fuel, accidental leakage can create serious safety risks, including fire and explosions. Therefore, continuous monitoring and timely detection of gas leakage are important for improving safety.

This project presents an LPG gas leakage detection system based on the STM32F446RE microcontroller and MQ-6 gas sensor. The MQ-6 sensor detects LPG and other flammable gases in the surrounding air and produces an electrical signal corresponding to the detected gas concentration. The analog signal is acquired through the Analog-to-Digital Converter (ADC) of the STM32 microcontroller.

The firmware processes the sensor readings and monitors changes relative to an initial baseline. The leakage decision is designed to consider the measured gas level, its trend over time, and the persistence of elevated readings. Based on the detected condition, the system provides three levels of visual indication using green, yellow, and red LEDs, representing Normal, Warning, and Critical conditions, respectively.

The prototype demonstrates how embedded systems can be used to monitor gas levels and provide local visual alerts. It also provides a foundation for future improvements, such as remote monitoring, data logging, and automatic gas shut-off.

1.1 PURPOSE OF THE SYSTEM

The primary purpose of this project is to design and implement an LPG leakage monitoring system using an MQ-6 gas sensor and an STM32 microcontroller.

The specific purposes of the system are:

- To detect changes in the surrounding LPG concentration using the MQ-6 gas sensor.
- To interface the sensor output with the STM32F446RE ADC for digital acquisition.
- To establish an initial sensor baseline under normal operating conditions.
- To monitor changes in sensor readings over time.
- To implement decision logic based on gas level, trend, and persistence.
- To indicate the detected condition using three LEDs.
- To provide a low-cost prototype for studying embedded gas detection and monitoring.

The system aims to improve awareness of potential LPG leakage through continuous electronic monitoring rather than relying entirely on human observation. The prototype is intended for educational and experimental use and is not a replacement for a certified gas safety alarm.

1.2 RELEVANCE AND SCOPE OF THE SYSTEM

LPG leakage detection is an important safety application because leaked gas can accumulate in poorly ventilated areas and may ignite when exposed to an ignition source. A microcontroller-based monitoring system provides a practical way to acquire sensor data, process readings, and communicate the detected condition.

The proposed system combines a gas sensor with an embedded controller to perform continuous monitoring and visual indication. The use of the STM32F446RE allows the system to implement ADC-based acquisition and programmable decision logic.

The scope of the project includes:

- Detection of LPG-related changes using the MQ-6 sensor.
- Acquisition of analog sensor output using the STM32 ADC.
- Baseline calibration and monitoring of sensor readings.
- Analysis of gas level, trend, and persistence.
- Three-level status indication using green, yellow, and red LEDs.
- Experimental observation of sensor response under different test conditions.

The current prototype provides local LED indication only. It does not include a buzzer, wireless communication, an LCD, or an automatic gas shut-off mechanism. Its performance depends on sensor placement, calibration, environmental conditions, and the decision thresholds implemented in the firmware.

Chapter 2

PROBLEM DEFINITION

LPG is commonly stored in cylinders and supplied through regulators and pipelines. Leakage can occur because of damaged hoses, loose connections, faulty regulators, or equipment deterioration. If leakage is not detected promptly, gas may accumulate and create hazardous conditions.

Manual inspection alone cannot continuously monitor the surrounding atmosphere. In addition, gas sensor readings can fluctuate because of environmental changes, sensor warm-up, and electrical noise. A detection system that responds to every temporary increase may produce false alarms, while an overly restrictive threshold may delay the detection of a genuine leak.

Therefore, a suitable embedded system should acquire sensor readings continuously, establish a reference baseline, evaluate changes in readings, and provide clear status indications based on defined decision criteria.

2.1 PROBLEM STATEMENT

The problem addressed by this project is the development of a low-cost embedded system capable of detecting potential LPG leakage using an MQ-6 gas sensor and an STM32F446RE microcontroller.

The system must interface the sensor with the microcontroller ADC, establish a baseline under normal conditions, and analyse changes in sensor readings over time. The decision logic should consider the measured gas level, the direction and rate of change in readings, and whether an elevated reading persists for a defined period.

The system must classify the monitored condition into three categories: Normal, Warning, and Critical. These categories are indicated using green, yellow, and red LEDs, respectively.

The central challenge is to provide meaningful visual indication while reducing the effect of temporary sensor fluctuations. Since the prototype is based on a low-cost gas sensor, its readings must be interpreted carefully and should not be treated as certified measurements of LPG concentration without appropriate calibration.

2.2 OBJECTIVES

The main objective is to design and implement an STM32-based LPG leakage detection prototype with three-level visual indication.

The specific objectives are:

1. To design an LPG leakage detection system using the STM32F446RE microcontroller.
2. To develop an interface between the MQ-6 gas sensor and the STM32 ADC.
3. To implement sensor calibration and baseline monitoring.
4. To develop leakage decision logic based on gas level, trend, and persistence.
5. To control three LEDs to indicate Normal, Warning, and Critical conditions.
6. To evaluate the sensor response and observe false-alarm behaviour during experimental testing.

Chapter 3

REQUIREMENT ANALYSIS

Requirement analysis identifies the hardware, software, and operational requirements needed to design and implement the proposed LPG leakage detection system. The system consists of an MQ-6 gas sensor, STM32F446RE microcontroller, three LEDs, resistors, a breadboard, jumper wires, and a USB power/programming connection.

The hardware detects and processes gas-related sensor changes, while the firmware performs sensor acquisition, baseline monitoring, decision-making, and LED control.

3.1 HARDWARE COMPONENTS REQUIRED

The following hardware components are used in the prototype.

Sl. No.	Component	Specification	Purpose
1	STM32 Nucleo-F446RE	STM32F446RE, ARM Cortex-M4, up to 180 MHz, 12-bit ADC	Processes sensor data and controls outputs
2	MQ-6 gas sensor	LPG/flammable gas sensor module, nominal 5 V supply	Detects changes in gas concentration
3	Green LED	5 mm	Indicates Normal condition
4	Yellow LED	5 mm	Indicates Warning condition
5	Red LED	5 mm	Indicates Critical condition
6	LED resistors	220 Ω	Limit LED current
7	Voltage-divider resistor(s)	10 kΩ	Reduce the sensor output voltage to a suitable ADC input level

Sl. No.	Component	Specification	Purpose
8	Breadboard	Solderless prototyping board	Supports temporary circuit assembly
9	Jumper wires	Male-to-male and/or male-to-female, as required	Connects the circuit components
10	USB cable	Compatible with the Nucleo board	Powers the board and supports programming/debugging

STM32 Nucleo-F446RE: The STM32 board serves as the central processing unit. Its ADC converts the analog sensor signal into a digital value that can be analysed by the firmware. Its GPIO pins control the three status LEDs.

MQ-6 gas sensor: The MQ-6 is a semiconductor gas sensor commonly used for detecting LPG and other flammable gases. Its sensing element changes electrical resistance in response to gas exposure. The sensor module converts this behaviour into an electrical output. The sensor requires a suitable supply voltage and a warm-up period.

LED indicators: Three 5 mm LEDs provide visual feedback. The green LED represents Normal, the yellow LED represents Warning, and the red LED represents Critical conditions.

Resistors: The 220 Ω resistors limit the current through the LEDs. The 10 kΩ resistor or resistor network is used in the sensor interface voltage divider, according to the implemented circuit.

Breadboard and jumper wires: These components enable the prototype to be assembled and modified without soldering a permanent PCB.

Power supply: The Nucleo board is powered through USB. The MQ-6 module requires its specified supply voltage, nominally 5 V for the module used. The sensor interface must ensure that the STM32 ADC input remains within its permitted voltage range.

3.2 SOFTWARE/FIRMWARE TOOLS USED

The software environment provides the tools required to configure the microcontroller, develop the firmware, program the board, and observe the system's behaviour.

STM32CubeIDE: STM32CubeIDE is the integrated development environment used for STM32 application development. It supports project configuration, C programming, compilation, debugging, and firmware flashing.

Embedded C: Embedded C is used to implement the application logic. The firmware reads the ADC, processes sensor values, evaluates the detected condition, and controls the LEDs.

STM32 HAL libraries: The Hardware Abstraction Layer provides functions for configuring and accessing STM32 peripherals, including ADC and GPIO.

ADC peripheral: The ADC converts the analog sensor voltage into a digital value. The STM32F446RE provides a 12-bit ADC, producing nominal digital values from 0 to 4095 when configured for 12-bit conversion.

GPIO peripheral: General-Purpose Input/Output pins are configured as outputs to control the green, yellow, and red LEDs.

The general firmware sequence is:

1. Initialize the required peripherals.
2. Allow the sensor to warm up and stabilize.
3. Acquire initial readings and establish a baseline.
4. Read the sensor output repeatedly.
5. Process readings and evaluate changes over time.
6. Classify the current condition.
7. Set the appropriate LED output.
8. Repeat the monitoring process.

3.3 FEASIBILITY ANALYSIS

The feasibility of the proposed system is evaluated in terms of technical, economic, and operational factors.

1. Technical feasibility

The STM32F446RE provides ADC and GPIO peripherals suitable for sensor acquisition and LED control. The MQ-6 sensor provides an electrical output that can be interfaced with the microcontroller through an appropriately designed voltage divider. The required firmware operations are achievable using Embedded C and STM32 HAL libraries.

However, successful integration depends on correct voltage interfacing, sensor stabilization, suitable decision thresholds, and appropriate grounding.

2. Economic feasibility

The system uses commonly available electronic components and a development board.

The breadboard-based construction avoids the initial expense of manufacturing a custom PCB. The 5 mm LEDs and resistors are inexpensive components, making the prototype suitable for an academic project.

Item Name	Amount (INR)
POT METER 16MM 10K	15.25
MQ-6 GAS SENSOR MODULE	92.37
LED 5MM RED	1.70
LED 5MM YELLOW	1.70
LED 5MM GREEN	1.70
BUZZER MINI 5V	12.71
220E Resistor (5 pcs)	4.24
10K Resistor (5 pcs)	4.24
GRAND TOTAL	**158.00**

3. Operational feasibility

The system is relatively simple to operate. Once powered and initialized, it continuously acquires sensor readings and indicates the classified condition through LEDs. No continuous user interaction is required during normal monitoring.

The system nevertheless requires suitable sensor placement, an adequate warm-up period, periodic inspection, and calibration. It is a prototype and should not be used as the sole protection against LPG leakage.

Chapter 4

SYSTEM DESIGN

The system design describes the hardware architecture, circuit connections, component arrangement, and firmware operation. The proposed architecture integrates the MQ-6 sensor with the STM32F446RE and uses three LEDs to communicate the detected condition.

The design has two main parts: the hardware subsystem, which acquires and indicates the sensor state, and the firmware subsystem, which processes sensor readings and determines the LED output.

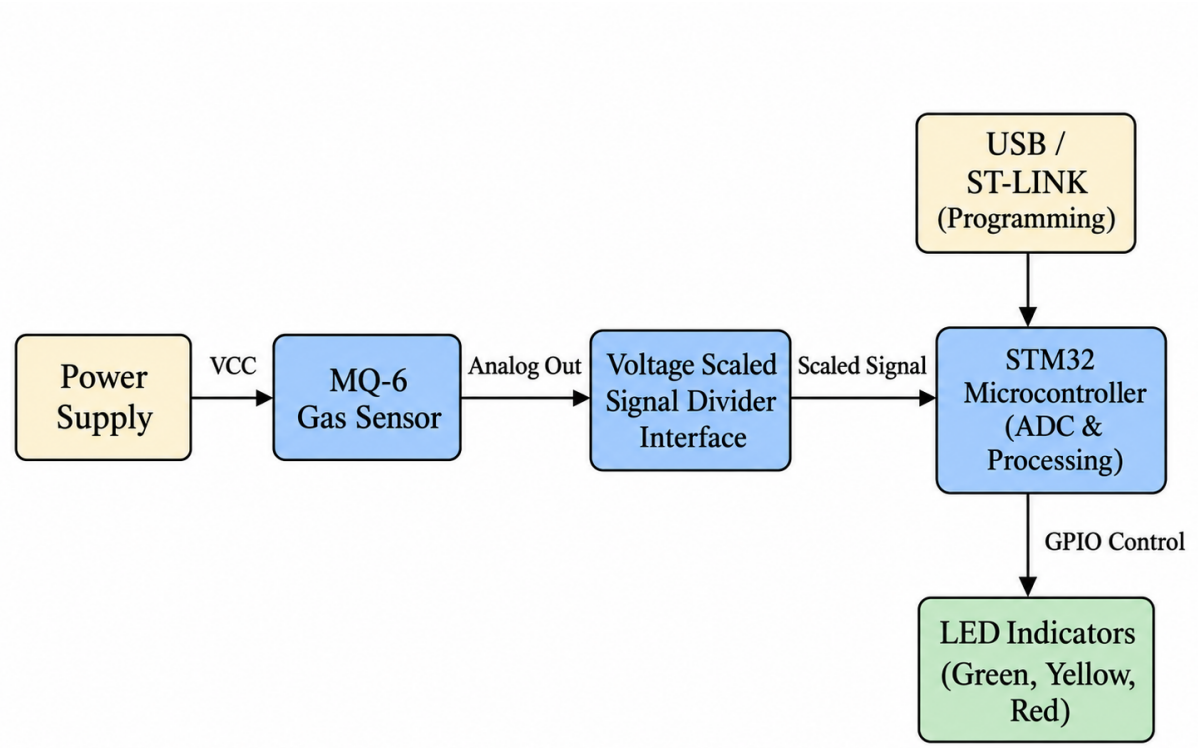
4.1 BLOCK DIAGRAM OF THE PROPOSED HARDWARE SYSTEM

The proposed hardware consists of a power supply, MQ-6 gas sensor, STM32 microcontroller, voltage-divider interface, and three LED indicators.

The MQ-6 sensor monitors the surrounding air and generates an analog output related to its response to flammable gases. The signal passes through the voltage-divider interface before reaching the STM32 ADC input. The microcontroller converts the signal into digital readings and processes the readings using the implemented firmware.

Based on the decision logic, the microcontroller activates the appropriate LED.

The main signal flow is:



The three output states are:

- Green LED: Normal condition.
- Yellow LED: Warning condition.
- Red LED: Critical condition.

The power supply supports the microcontroller and sensor according to their electrical requirements. The USB connection also allows firmware programming and debugging through the Nucleo board's onboard ST-LINK interface.

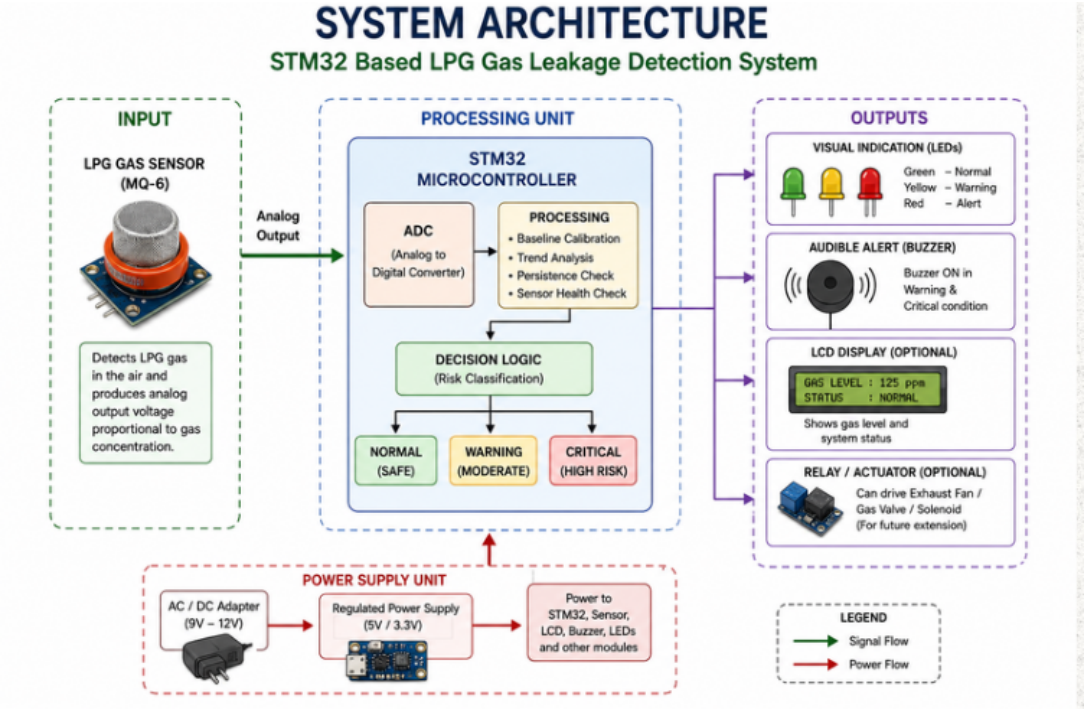


Figure 4.1: System architecture of the STM32-based LPG leakage detection system.

4.2 CIRCUIT SCHEMATIC / WIRING DIAGRAMS

The circuit connects the MQ-6 sensor to the STM32 ADC and the three LEDs to GPIO output pins.

The following table represents the intended connection arrangement.

Circuit Connections

Component	Pin	Goes to
MQ-6	VCC	5 V
MQ-6	GND	GND
MQ-6	AO	10 kΩ divider → PA0
MQ-6	DO	Not connected
10 kΩ #1	One end	MQ-6 AO
10 kΩ #1	Other end	Divider junction
10 kΩ #2	One end	Divider junction
10 kΩ #2	Other end	GND
Green LED	Anode	PB0 → 220 Ω
Green LED	Cathode	GND
Yellow LED	Anode	PB1 → 220 Ω
Yellow LED	Cathode	GND
Red LED	Anode	PB2 → 220 Ω
Red LED	Cathode	GND

Sensor interface: MQ-6 AO → voltage divider → STM32 PA0 (ADC1)

The voltage-divider circuit must be designed to keep the ADC input within the STM32's allowable input voltage. For a simple two-resistor divider,

$$V_{\text{ADC}} = V_{\text{in}} \frac{R_2}{R_1 + R_2}$$

where R_1 is connected between the sensor output and the divider junction, and R_2 is connected between the junction and ground.

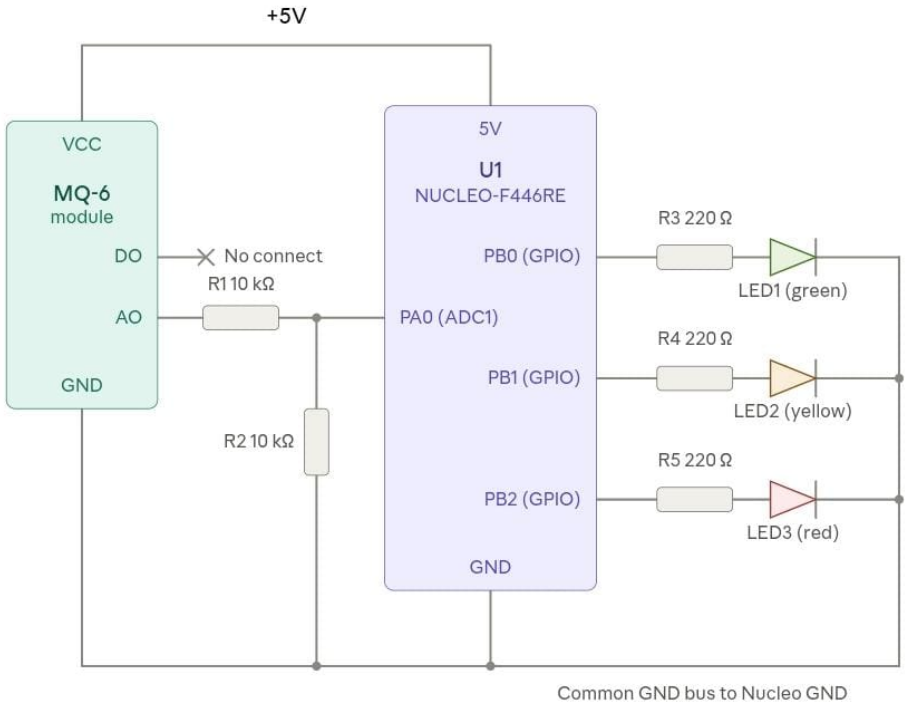


Figure 4.2: Circuit schematic of the LPG leakage detection prototype.

4.3 COMPONENT LAYOUT OR PCB DESIGN

The prototype is assembled on a solderless breadboard rather than a custom PCB. The breadboard provides a convenient platform for connecting the sensor, microcontroller signals, LEDs, and resistors during development.

The Nucleo board acts as the main controller and is connected to the breadboard using jumper wires. The MQ-6 sensor is placed where it can sample the surrounding air. The LEDs are arranged so that their colours can be observed easily during testing.

The 220 Ω resistors are connected in series with the LED circuits to limit current. The voltage-divider components are placed between the MQ-6 analog output and the ADC input.

During assembly, the following layout considerations are important:

- Maintain clear separation between the sensor supply and the ADC signal connection.
- Ensure that the sensor output is not connected directly to an ADC input if its voltage could exceed the STM32's permitted range.
- Establish a common ground where required by the circuit.
- Avoid loose jumper-wire connections.
- Place the LEDs so their status indications are easily visible.

- Keep the sensor exposed to the surrounding air rather than obstructing its sensing surface.

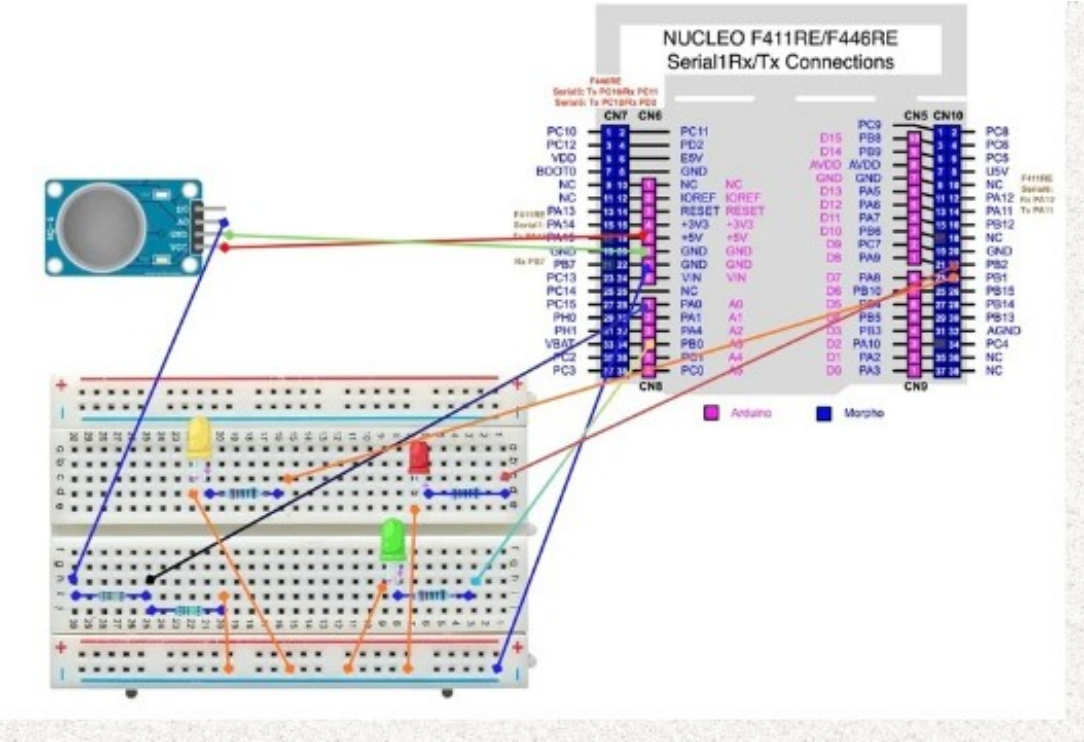


Figure 4.3: Component layout of the hardware prototype.

4.4 WORKING PRINCIPLE AND FLOWCHARTS

The system operates by acquiring analog sensor readings, comparing them with a baseline, and evaluating changes over time.

Step 1: Sensor initialization and warm-up

After power is applied, the MQ-6 sensor is allowed to warm up and stabilize. The microcontroller initializes the ADC and GPIO peripherals.

Step 2: Baseline calibration

The system acquires initial sensor readings under the selected normal test condition. These readings are used to establish a reference baseline. The baseline represents the sensor's observed response during calibration and is not automatically equivalent to zero gas concentration.

Step 3: Sensor data acquisition

The ADC repeatedly converts the sensor voltage into digital values. Multiple samples may be averaged to reduce the influence of short-term fluctuations.

Step 4: Signal analysis

The firmware evaluates the current readings in relation to the baseline. It may consider the magnitude of the change, whether the readings are increasing, and whether the elevated readings persist over time.

Step 5: Condition classification

The processed reading is compared with the decision criteria implemented in the firmware. The system classifies the condition as Normal, Warning, or Critical.

Step 6: LED indication

The microcontroller updates the three LED outputs according to the classification. The intended indication is green for Normal, yellow for Warning, and red for Critical.

Step 7: Continuous monitoring

The process repeats so that the system can respond to subsequent changes in the sensor readings.

The general flowchart is:

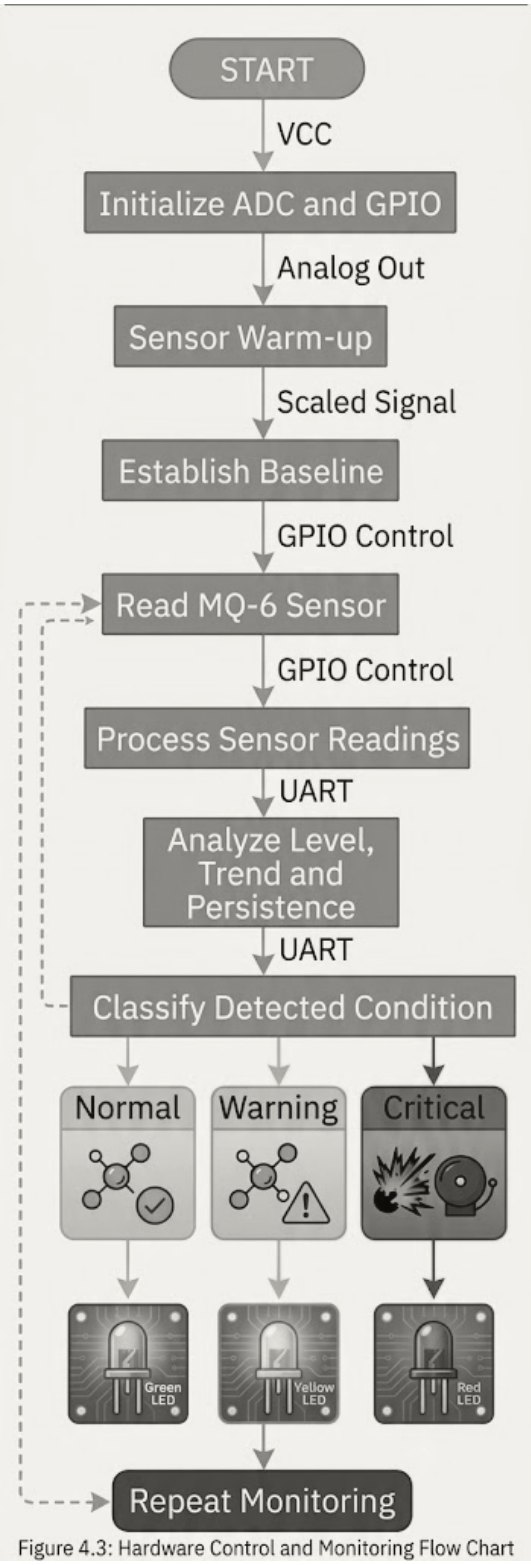


Figure 4.4: Flow Chart of the Proposed Hardware System

Chapter 5

IMPLEMENTATION

The implementation phase involves assembling the hardware, establishing the sensor interface, configuring the STM32 peripherals, developing the firmware, and testing the resulting prototype.

5.1 STEP-BY-STEP ASSEMBLY OF THE HARDWARE PROTOTYPE

The hardware prototype was assembled using the STM32 Nucleo-F446RE board, MQ-6 gas sensor, three 5 mm LEDs, resistors, a breadboard, jumper wires, and a USB cable.

The assembly procedure is as follows:

1. Prepare the development board: Place the STM32 Nucleo-F446RE board on a stable surface and identify the required power, ground, ADC, and GPIO pins.
2. Connect the gas sensor: Connect the MQ-6 module to its specified power supply and ground.
3. Construct the voltage divider: Connect the sensor analog output to the divider circuit and route the divided output to the selected STM32 ADC input.
4. Connect the LEDs: Connect the green, yellow, and red LEDs to their designated GPIO pins through individual 220 Ω current-limiting resistors.
5. Connect the ground: Ensure that the relevant circuit grounds are connected correctly.
6. Inspect the wiring: Verify the polarity of the LEDs, resistor placement, supply connections, and ADC input voltage range.
7. Connect USB: Connect the Nucleo board to a computer using a compatible USB cable for power, programming, and debugging.
8. Run preliminary checks: Confirm that the microcontroller powers up and that the

firmware can control the LEDs.

9. Perform sensor testing: Observe the sensor readings under baseline conditions and during controlled gas-sensor response tests.

5.2 CIRCUIT INTEGRATION AND WIRING DETAILS

Circuit integration connects the sensor interface, microcontroller, and output indicators into one functional system.

The MQ-6 sensor requires its specified supply voltage. Its analog output is connected to the STM32 through the voltage divider so that the ADC input stays within the permitted voltage range. The STM32 ADC reads this signal, while GPIO pins provide the output control for the three LEDs.

Each LED is connected in series with a 220 Ω resistor. The resistor limits current and helps protect the LED and microcontroller output. The LED polarity must be checked during wiring.

The 10 kΩ resistor or resistor network is used in the voltage-divider interface according to the actual circuit design. The precise voltage-divider output depends on the complete resistor arrangement, so the wiring should be documented using the final circuit schematic.

Before the system is operated, the following was completed:

- Verified the sensor supply voltage.
- Confirmed the common ground connections.
- Checked the voltage-divider output.
- Verified that the ADC input does not exceed the permitted range.
- Confirmed the GPIO-to-LED connections.
- Checked for short circuits and loose wires.

5.3 FIRMWARE/CODE FLASHING AND CONFIGURATION

The firmware is developed using STM32CubeIDE and Embedded C. The STM32 HAL libraries are used to configure the ADC and GPIO peripherals.

The implementation process consists of the following stages:

Project configuration: Create an STM32 project for the Nucleo-F446RE board and configure the required peripherals.

GPIO configuration: Configure the selected LED pins as digital outputs.

ADC configuration: Configure the ADC channel connected to the sensor interface. The conversion settings must match the application requirements.

Firmware development: Implement sensor acquisition, baseline calibration, signal processing, decision logic, and LED control.

Compilation: Build the project and resolve any compiler errors or configuration issues.

Firmware flashing: Connect the board through USB and use the onboard ST-LINK interface to program the compiled firmware.

Debugging and monitoring: Use debugging tools or serial output, if configured, to inspect ADC readings and verify the decision logic.

The high-level firmware logic is:

```
Initialize_ADC();
Initialize_GPIO();
Initialize_UART();
Wait_For_Sensor_Stabilization();
Establish_Baseline();

while (1)
{
    Read_Sensor_ADC();
    Process_Readings();
    Evaluate_Gas_Level_Trend_And_Persistence();
    Classify_Condition();
    Update_Status_LEDs();
    Send_Status_Through_UART();
}
```

5.4 CHALLENGES FACED DURING PHYSICAL ASSEMBLY AND HOW THEY WERE FIXED

Sensor voltage compatibility: The MQ-6 module may operate at 5 V, while the STM32 uses 3.3 V logic. The analog output must therefore be checked and appropriately scaled before it is applied to the ADC.

Sensor reading fluctuations: Sensor readings may vary during warm-up or because of environmental changes. Baseline monitoring and averaging can help reduce the influence of short-term fluctuations.

LED connection errors: Incorrect LED polarity or missing current-limiting resistors can prevent correct operation. The connections are checked against the circuit diagram.

ADC configuration: Incorrect channel selection or peripheral configuration can cause readings to remain fixed or differ from expectations. The ADC settings and wiring should be verified together.

Loose breadboard connections: Poor contact between jumper wires and breadboard terminals can produce intermittent behaviour. The wiring should be inspected and reconnected where necessary.

Decision-threshold selection: Thresholds that are too sensitive may cause false warnings, while thresholds that are too restrictive may delay detection. Controlled testing and documented observations are needed to select suitable values.

5.5 PHOTOGRAPHS OF THE BUILT HARDWARE MODEL

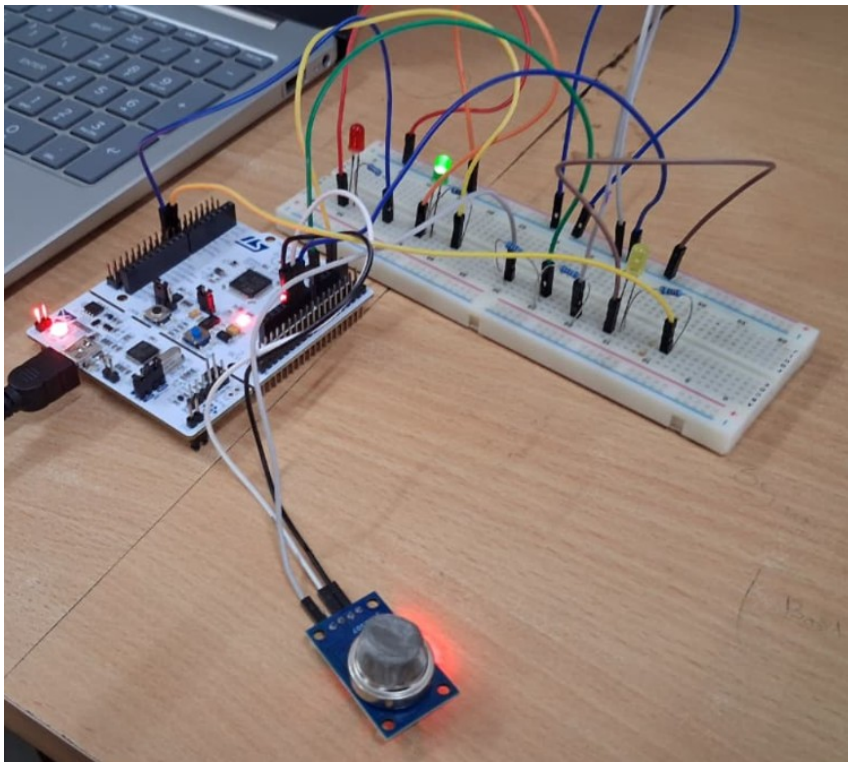


Figure 5.1: Output during normal environmental conditions

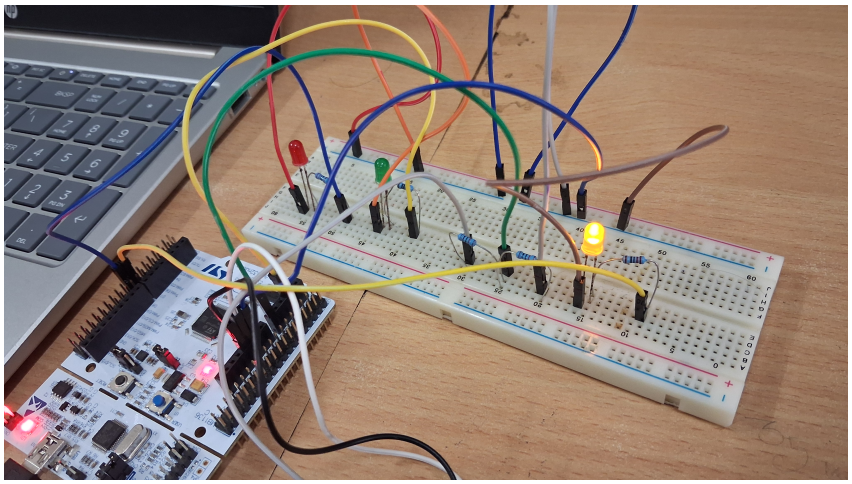


Figure 5.2: Output during warning environmental conditions

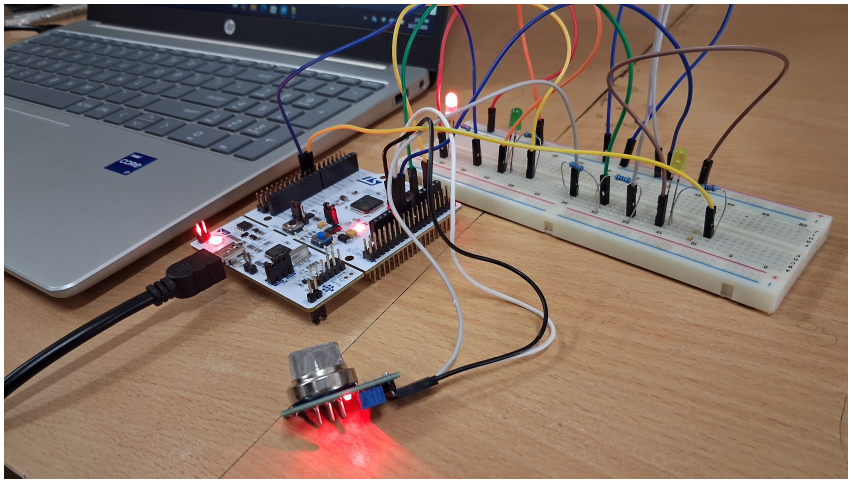


Figure 5.3: Output during critical environmental conditions

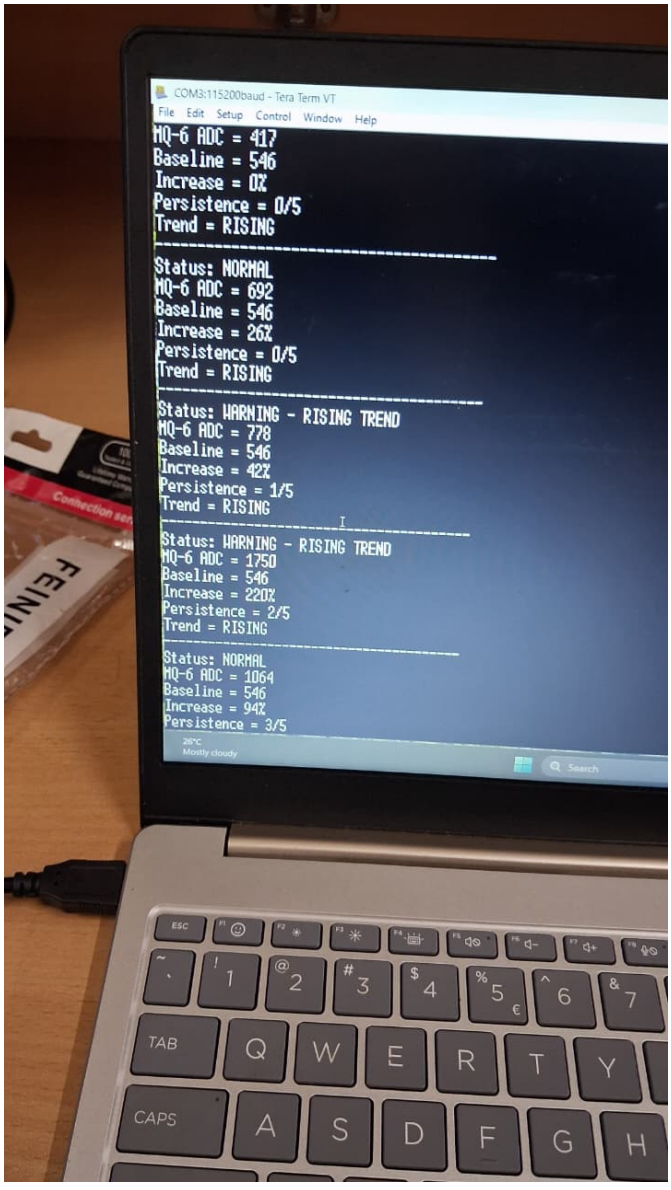


Figure 5.4: Prototype during sensor response testing.

Chapter 6

TESTING, RESULTS, AND PERFORMANCE ANALYSIS

Testing is required to verify that the assembled hardware and firmware behave as intended. The system should be evaluated under baseline conditions, during a controlled increase in sensor response, and after the stimulus is removed.

The testing focuses on ADC acquisition, baseline stability, LED operation, response to changing sensor readings, and the behaviour of the decision logic. Quantitative conclusions should be based on recorded measurements rather than assumed performance.

6.1 TESTING PROCEDURES

The following procedures can be used to evaluate the prototype.

Power supply verification

Check that the Nucleo board and MQ-6 sensor receive their specified supply voltages. Confirm that the grounds are connected correctly and that the ADC input remains within its permitted range.

Sensor stabilization

Allow the MQ-6 sensor to warm up according to the module's specifications. Observe the readings until they are sufficiently stable for the chosen baseline procedure.

Baseline calibration

Record the ADC readings under the selected normal test conditions. Calculate the baseline using the method implemented in the firmware, such as an average of multiple initial

readings.

Normal-condition testing

Observe the sensor readings while the sensor is exposed to the baseline environment. Confirm that the system indicates the Normal state when the decision criteria are satisfied.

Sensor-response testing

Under an appropriately controlled and safe test procedure, observe changes in sensor readings when the sensor response increases. Record the ADC values and corresponding LED states.

Recovery testing

After the test stimulus is removed, continue recording readings to observe whether the sensor response returns toward its previous baseline.

False-alarm observation

Repeat the baseline test and examine whether temporary fluctuations cause Warning or Critical indications. Record the number of observed false activations and the test duration.

Because LPG is flammable, testing should not involve deliberate release of large quantities of gas, ignition sources, or unsafe indoor accumulation. A controlled laboratory procedure and suitable ventilation should be used.

6.2 ACTUAL SYSTEM READINGS, OUTPUT DATA, AND PERFORMANCE GRAPHS

The system output was evaluated using the ADC readings obtained from the MQ-6 sensor, the calibrated clean-air baseline, the percentage increase from the baseline, the persistence count, the detected trend, and the corresponding LED status. During the experiment, the calibrated baseline ADC value was 546.

Test Condition	**ADC Reading**	**Baseline**	**LED Output**	**Observation**
Initial stabilization	–	–	Green	Sensor stabilization before baseline calibration.
Normal environment	417–692	546	Green	ADC remained at or near the clean-air reference.

Increased sensor response	778	546	Yellow	42% increase with a rising trend was detected.
Sustained elevated response	1689–2059	546	Yellow → Red	Elevated response persisted and resulted in warning followed by critical classification.
Recovery after stimulus removal	983 → 800	546	Red → Yellow	ADC decreased from the peak value, indicating recovery from the elevated response.

The calibrated baseline was obtained by averaging the clean-air sensor readings after the stabilization period. The measured baseline was **546 ADC**. During the gas exposure test, the ADC value increased significantly above the baseline. The highest recorded value was **2059 ADC**, corresponding to a displayed increase of **277%** over the baseline. The system subsequently classified the condition as critical and activated the red LED.

After removal of the gas stimulus, the ADC value decreased from the peak, with readings such as 983 and 800 ADC being observed. This demonstrated the sensor's response and recovery behaviour, although these values were still above the clean-air baseline.

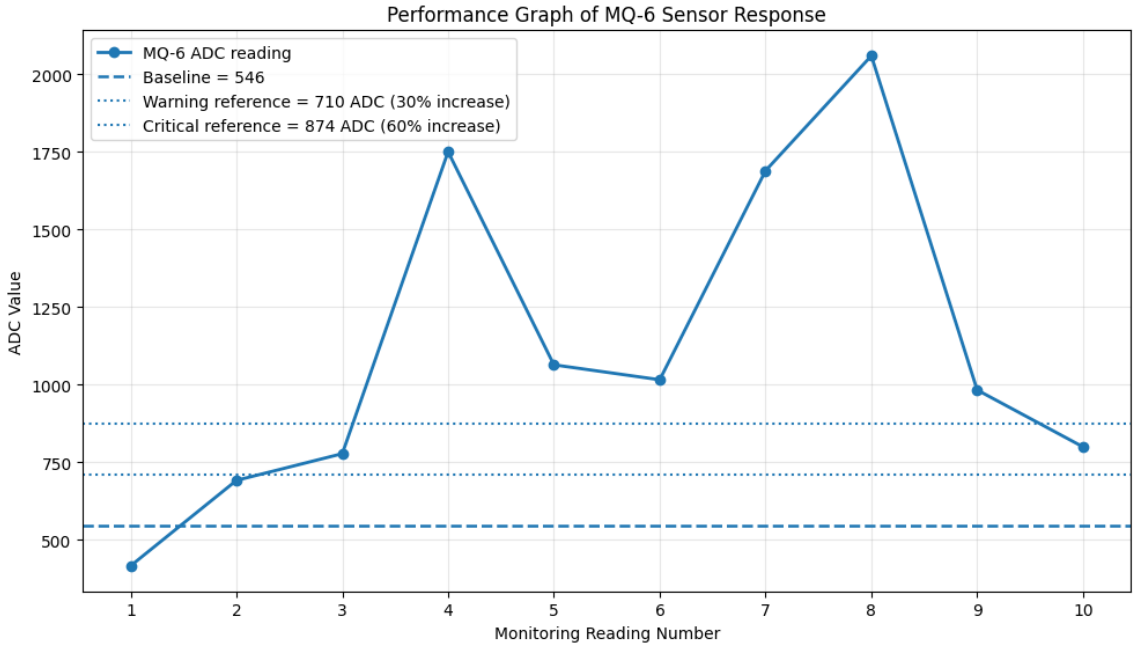


Figure 6.1: Performance graph of MQ-6 sensor response with respect to the calibrated baseline

Chapter 7

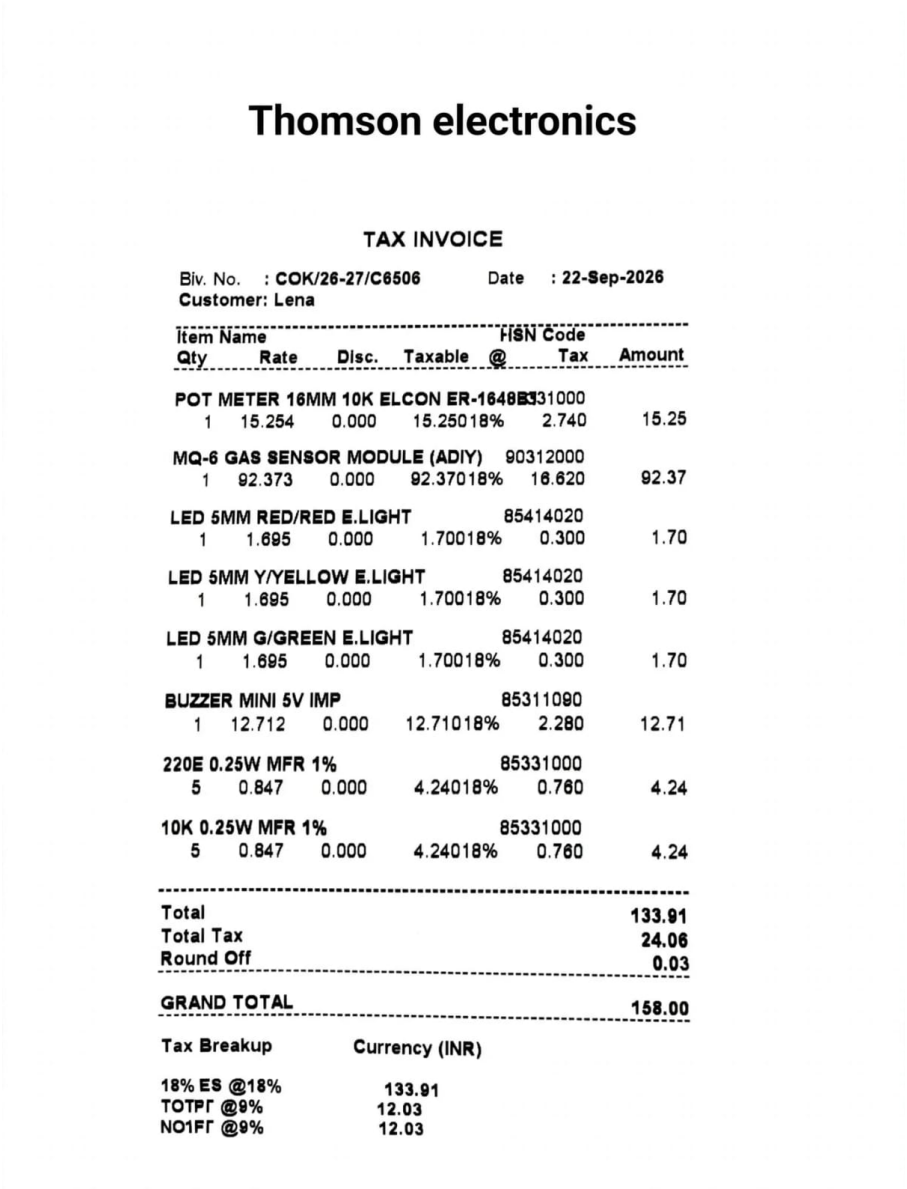
COST ESTIMATION AND FINANCIAL ANALYSIS

The financial analysis estimates the cost of developing the prototype from its constituent hardware components. It also considers the difference between an educational prototype and a commercially manufactured LPG detector.

The final cost depends on the purchase price of the development board, sensor module, passive components, and prototyping materials. The cost table should therefore be completed using actual purchase prices or documented supplier quotations.

7.1 BILL OF MATERIALS (BOM) WITH COMPONENT PRICES

The bill of materials lists the components used to assemble the system.



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TAX INVOICE

Biv. No. : COK/26-27/C6506 Date : 22-Sep-2026
Customer: Lena

Item Name	Qty	Rate	Disc.	Taxable	HSN Code @	Tax	Amount
POT METER 16MM 10K ELCON ER-1648E3	1	15.254	0.000	15.250	85331000 18%	2.740	15.25
MQ-6 GAS SENSOR MODULE (ADIY)	1	92.373	0.000	92.370	90312000 18%	16.620	92.37
LED 5MM RED/RED E.LIGHT	1	1.695	0.000	1.700	85414020 18%	0.300	1.70
LED 5MM Y/YELLOW E.LIGHT	1	1.695	0.000	1.700	85414020 18%	0.300	1.70
LED 5MM G/GREEN E.LIGHT	1	1.695	0.000	1.700	85414020 18%	0.300	1.70
BUZZER MINI 5V IMP	1	12.712	0.000	12.710	85311090 18%	2.280	12.71
220E 0.25W MFR 1%	5	0.847	0.000	4.240	85331000 18%	0.760	4.24
10K 0.25W MFR 1%	5	0.847	0.000	4.240	85331000 18%	0.760	4.24

Total	133.91
Total Tax	24.06
Round Off	0.03
GRAND TOTAL	158.00

Tax Breakup	Currency (INR)
18% ES @18%	133.91
TOTPΓ @9%	12.03
NO1FΓ @9%	12.03

The total cost is Rs. 2158 [including STM32 and USB]

7.2 TOTAL DEVELOPMENT COST VS. COMMERCIAL ALTERNATIVES

The proposed system is an educational prototype designed to demonstrate gas sensing, ADC acquisition, embedded decision logic, and visual indication.

A commercial LPG detector may include a calibrated sensing element, an audible alarm, an enclosure, dedicated power circuitry, manufacturing quality controls, and certification appropriate to its intended application.

The comparison should consider more than purchase price.

Parameter	Proposed Prototype	Commercial LPG Detector
Primary purpose	Educational and experimental monitoring	Product-specific gas safety monitoring
Controller	STM32F446RE development board	Depends on the product
Sensor	MQ-6 gas sensor module	Depends on the product
Indication	Three LEDs	Depends on the product; may include audible alerts
Calibration	Prototype calibration procedure	Depends on manufacturer specifications
Enclosure	Breadboard-based prototype	Typically product-specific enclosure
Certification	Not certified as a safety device	Depends on the particular model and applicable standards
Cost	Calculated from the project BOM	Based on a documented product price

Chapter 8

CONCLUSION AND FUTURE SCOPE

The project demonstrates the design and implementation of an LPG leakage monitoring prototype using the STM32F446RE microcontroller and MQ-6 gas sensor. The system acquires the sensor output through the ADC and processes the readings to determine the detected condition.

The firmware architecture supports baseline monitoring and decision-making based on gas level, trend, and persistence. Three LEDs provide local visual indication of Normal, Warning, and Critical states.

The prototype provides a practical platform for learning about embedded system development, analog sensor interfacing, digital signal acquisition, firmware design, and hardware integration. Further testing and calibration are required to establish its detection performance and suitability for any specific operating environment.

8.1 SUMMARY OF ACHIEVEMENTS AND PROJECT IMPACT

The main achievements of the project are:

- Integration of an MQ-6 gas sensor with the STM32F446RE microcontroller.
- Acquisition of the analog sensor signal through the STM32 ADC.
- Development of a baseline-monitoring approach for sensor readings.
- Design of decision logic that considers gas level, trend, and persistence.
- Implementation of three-level visual indication using green, yellow, and red LEDs.
- Assembly of a low-cost breadboard prototype for experimental evaluation.

The project illustrates how a microcontroller can be used to convert sensor signals into meaningful status indications. It also provides a foundation for further work in gas monitoring and embedded safety applications.

8.2 CURRENT LIMITATIONS OF THE PROTOTYPE

The current prototype has several limitations.

Sensor calibration: The MQ-6 output depends on sensor characteristics, operating conditions, and the surrounding environment. A baseline reading does not by itself provide an accurate measurement of LPG concentration.

Detection coverage: The system monitors the sensor's immediate environment. Gas distribution depends on leakage rate, airflow, ventilation, temperature, and sensor placement. Therefore, no universal detection distance can be assigned to the prototype without testing.

Power consumption: The MQ-6 uses a heated sensing element, which consumes a significant portion of the system's power. This makes the prototype less suitable for low-power battery operation without further redesign.

Warm-up time: The sensor requires a warm-up period, and its readings may drift during stabilization.

Environmental sensitivity: Temperature, humidity, other gases, and contamination may affect the sensor response.

Local indication only: The system uses three LEDs and does not include remote notifications, a display, or an audible alarm.

Safety certification: The prototype has not been established as a certified LPG safety device and must not be relied upon as the sole means of gas-leak protection.

8.3 RECOMMENDATIONS AND FUTURE UPGRADE POSSIBILITIES

The system can be improved in several ways.

Improved calibration: Develop a more systematic calibration procedure and evaluate the sensor response under controlled conditions. If quantitative gas concentration is required,

appropriate reference equipment and a validated calibration method will be necessary.

Display integration: Add an LCD or OLED display to show ADC readings, sensor status, and diagnostic information.

Remote monitoring: Add Wi-Fi or another suitable communication interface to transmit status information to a monitoring system.

Data logging: Record sensor readings and timestamps to support trend analysis and evaluation of system performance.

Improved decision logic: Evaluate averaging, filtering, hysteresis, and persistence-based rules using repeated tests to reduce false alarms without introducing excessive delay.

Power optimization: Investigate power requirements and possible alternatives if battery operation is required. Any reduction in sensor-heater power must be evaluated against the sensor manufacturer's operating requirements.

Automatic gas shut-off: A suitable gas-rated valve and dedicated control circuitry could be investigated as a future extension. Such a feature would require fail-safe design and thorough safety validation.

Enclosure and PCB development: Replace the breadboard arrangement with a designed PCB and suitable enclosure to improve mechanical stability and repeatability.

These upgrades could improve usability and monitoring capabilities. Any application intended to protect people or property would also require appropriate safety engineering, validation, and compliance with applicable standards.

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